logisth 0.4.0 — User Manual



A two-part polyphonic subtractive synthesizer for VST3, CLAP and standalone use. This manual describes logisth's own controls and behaviour. logisth is an independent instrument and is not affiliated with LennarDigital. It does not load Sylenth1 presets or reproduce its sound exactly.

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1. Installation and first sound

Download the archive for your operating system from the GitHub release. Each native archive contains VST3, CLAP and Standalone folders. Extract it first. Copy the entire .vst3 bundle and the .clap file or bundle into a plug-in directory scanned by your DAW. Keep the internal structure of bundles intact.

System	User VST3 directory	User CLAP directory
Windows	%LOCALAPPDATA%\Programs\Common\VST3	%LOCALAPPDATA%\Programs\Common\CLAP
macOS	~/Library/Audio/Plug-Ins/VST3	~/Library/Audio/Plug-Ins/CLAP
Linux	~/.vst3	~/.clap

If your DAW does not scan these user directories, add them to its scan paths or use its documented system directory. Rescan plug-ins, insert logisth on a MIDI instrument track, select Init / analog from SOUND LIBRARY, and play a note. The on-screen keyboard also generates notes. Start at a moderate listening level.

For Standalone, launch the executable or app and choose audio and MIDI devices in its audio settings. The standalone wrapper provides its own device settings; those are separate from the synth panel.

Windows builds target x64. macOS Universal bundles contain Intel x86_64 and Apple Silicon arm64 code and require macOS 11 or later. The minimum Linux glibc version is recorded in BUILD_INFO.json. macOS bundles have an ad-hoc signature, without Apple notarization; Windows binaries have no publisher signature. Follow your OS's normal review/open procedure for downloaded applications you trust.

VST2 is not included: building it requires a separately licensed VST2 SDK.

2. The single-window console

All sections occupy one console. No page needs vertical scrolling. Select A or B at the top to edit that part's two oscillators, filter and amplitude envelope. This selects the editing view; it does not mute the other part. Use SOLO for listening to one part. The central rack changes its controls when you select a rack name. The small square beside each name independently enables that section. Selecting a rack page does not enable or disable its processing.

The upper row contains the preset selector, previous/next preset buttons, A/B, SOLO, POLY, SYNC and THEME. Filters are on the left, the rack in the centre, and the mixer on the right. Mod Env, LFO and Misc routing remain visible below them. The bottom row contains wheels, the keyboard and mono/portamento controls.

Drag the bottom-right resize handle to scale the console from 80% to 150%. The nominal size is 1240 by 860 pixels. Initial size adapts to the display. THEME switches between LIGHT and DARK. Each instance saves its own theme with the plug-in state. The theme changes appearance without changing audio parameters.

Drag knobs and faders to adjust values. Numeric fields can be dragged or edited. Examples of accepted values include 2.5k for 2500 Hz, 350ms and 65%. Double-click a control to restore its default; modulation amount controls return to zero. Hover for a tooltip and the parameter/value readout at the bottom. Disabled phase knobs become available when their oscillator's RETRIG is enabled.

3. Oscillators and amplitude envelopes

There are four oscillators: A1 and A2 feed source A; B1 and B2 feed source B. Each oscillator has a waveform menu, 0–8 unison voices, volume, octave, note, fine tuning, phase, detune, stereo width, pan, retrigger and invert.

Control	Purpose
Wave	Selects the oscillator waveform.
Voices	Sets the unison voice count. Zero silences this oscillator.

Control	Purpose
Volume	Sets its level before filtering and the part envelope.
Octave / Note / Fine	Coarse octave, semitone and cent tuning.
Phase / Retrig	Retrig starts notes at the selected phase; otherwise phase runs without this reset.
Detune	Spreads unison pitches. Increase gradually for a wider ensemble.
Stereo	Spreads unison voices across the stereo field.
Pan	Positions the oscillator output left or right.
Invert	Inverts oscillator polarity; it can affect cancellation when sources mix.

The amplitude envelope for each part has attack, decay, sustain and release. Attack rises from silence, decay approaches sustain, and release fades after note-off. Sustain is a level; the other three are times. The line above the faders illustrates the current envelope. Amp Env A and B remain independent.

3.1 Four source engines

Every oscillator slot A1/A2/B1/B2 has its own engine menu: Classic, Wavetable, Sampler or Granular. All four may use different engines and different audio files. The adjacent view menu selects VOICE, SOURCE or ENV without opening another window. VOICE holds the shared tuning, level, unison, stereo and pan controls. SOURCE holds the engine-specific controls; ENV holds that oscillator's optional ADSR. The central Amp Env A/B remains the final envelope for its part.

Classic retains the original oscillator waveforms and retrigger behaviour. Wavetable scans up to 256 frames using POSITION. Frames are interpolated and precomputed spectral bands reduce aliasing at higher notes. WARP bends phase. Choose a frame size (256/512/1024/2048 samples) before IMPORT when loading a concatenated wavetable WAV. The factory table morphs between sine and harmonics. Wavetable extraction uses the file's left channel. Padding is used for files shorter than one frame; additional frames beyond 256 are ignored.

Sampler plays the imported audio at the selected ROOT MIDI note (default 60). START and END choose a region; LOOP repeats it with a short crossfade; REVERSE reverses playback. Without LOOP, playback ends at the selected boundary. Root sets the reference pitch, not automatic pitch detection. Stereo files retain both channels; mono files feed both channels. Unison and pitch bend still apply.

Granular overlaps up to eight Hann-windowed grains per unison voice. POSITION selects the source location; SIZE is grain duration in milliseconds; DENSITY is grains per second; SPRAY randomises locations around the playhead; SCAN moves through the material in cycles per second. Negative scan moves backwards. REVERSE changes each grain's playback direction. ROOT controls grain pitch. Unison, stereo and pan remain available in VOICE.

Enable OSC ENVELOPE in ENV to give a slot an additional attack, decay, sustain and release. A part's shared envelope still limits its final output. New Position A/B/A+B and Grain size A/B/A+B modulation destinations scan frames or sample locations and scale grain duration. Position changes are smoothed.

IMPORT accepts WAV, AIFF and FLAC at 8–96 kHz, up to 12 seconds per source. Alternatively, drop one file onto an oscillator panel: a Classic slot switches to Sampler, while the other engine selections are retained. Import decoding and table preparation run on a worker, and the footer reports completion or errors. RESET returns the slot's material to the factory glass sample/table. The waveform strip previews the loaded material. No original media path is needed on project recall: imported PCM data is compressed into the plug-in state, with a 64 MiB state-input limit. Save larger libraries separately; this is a single-sample source rather than a multisample instrument.

4. Filters and shared Filter Control

Each part has an input selector, type, slope, cutoff, resonance and drive. Input A uses A1+A2, B uses B1+B2, A+B uses both, and None disconnects the input. Consequently, a part's filter can process the other oscillator source. Its output still passes through its own amplitude envelope and Mix level.

Choose low-pass, high-pass, band-pass or bypass. Slope selects 12 or 24 dB per octave. Cutoff is the local frequency. Resonance emphasises the cutoff region; high resonance can self-oscillate while a note is active. Drive adds saturation.

The shared FILTER CONTROL affects both A and B while retaining their local settings. CUTOFF is an offset in semitones: +12 doubles both cutoff frequencies, -12 halves them. RESO adds a shared resonance offset. KEYTRACK ties cutoff to played pitch; at full tracking, an octave higher doubles cutoff. WARM changes the filter saturation characteristic. There is no third filter in this section. Modulation is combined with these settings before the effective cutoff is limited to a valid frequency for the current sample rate.

5. Modulation

Mod Env 1 and 2 each provide an ADSR envelope and two destination/amount pairs. LFO 1 and 2 also provide two pairs each. Misc 1 and Misc 2 each contain two source blocks, with two pairs per block. There are 16 routes in total. The source menu in a Misc block chooses the signal used by both routes in that block.

Choose a destination, then move its small amount knob away from zero. Amount ranges from -10 to +10; negative values reverse the modulation. None or amount zero makes a route inactive. Routes to the same destination are added together. The destination popup groups oscillators, filters and modulation/effects into three columns.

Destinations include Volume, Pitch, Phase and Pan for A, B or A+B; Cutoff and Resonance for A, B or A+B; Mix A, B or A+B; Phaser Frequency; LFO 1/2 Rate and Gain; and Distortion Amount. Modulating A+B affects both parts together.

Misc sources include Velocity, Keytrack, Aftertouch, Mod Wheel, Pitch Wheel, Amp Env A/B, Mod Env 1/2, LFO 1/2, Step Velocity and random sources. Polyphonic pressure takes precedence over channel pressure for the affected note. Random values are generated at note-on. Step Velocity follows the active arp step and returns to zero after its gate ends.

Each LFO has waveform, Rate, Gain, Offset and FREE. With SYNC enabled and FREE off, RATE is a rhythmic division at the host tempo. With FREE enabled it uses its Hz rate and does not restart on each note. Otherwise notes restart the LFO. Gain scales its output; Offset shifts it. The global fallback tempo is 120 BPM when no host tempo is available.

Modulation depth is specific to logisth. At amount +10 and source value +1, pitch adds 48 semitones, cutoff adds 120 semitones, phase adds one cycle, and pan adds one full side. Volume, Mix, Resonance, Gain and Distortion add one linear unit. LFO Rate multiplies by 16 and Phaser Frequency by 1024. Smaller amounts give smaller changes; start low. Effects are global, so FX modulation uses the newest active voice. LFO feedback modulation uses the previous sample.

6. Chords and pattern arpeggiator

The CHORDS rack page and Pattern arp use a chord-programming and multi-lane workflow inspired by Cthulhu, with logisth's own interface and implementation. It is not a Cthulhu clone and does not import Cthulhu chord/pattern files. The chain is MIDI input Chords Arpeggiator Synth and optional MIDI output. Each processor has its own enable square in the rack.

Chord mapper

Choose CHORDS and enable its square. TRIGGER selects one of 128 MIDI notes. VOICING assigns Unison, Major, Minor, Maj 7, Min 7, Dominant 7, Sus 2, Sus 4, Power, Diminished or Augmented to that trigger. MAP ALL assigns the selected type to all triggers, transposed to each root. Eight note fields edit the stored voicing directly: drag or enter MIDI numbers 0–127; -1 removes a voice. Empty triggers pass their original note through. Shared notes in overlapping chords are reference-counted so releasing one trigger does not cut another chord short.

LEARN collects up to eight incoming MIDI notes for the selected trigger. Play your voicing, then choose DONE. Notes are collected silently during learning; DONE stores the first eight distinct pitches in ascending order. The complete chord map is saved with the project. Chords can play as blocks with arp disabled, or supply a pool of notes to the Pattern

engine.

Pattern engine

Select ARPEGGIATOR, choose Pattern and enable its square. Use 01–16 / 17–32 to edit up to 32 steps. The grid has seven rows. The small field beside each row sets that lane's independent length from 1 to 32; TIE shares the GATE length.

Lane	Meaning
NOTE	Rank 1–8 in the sorted held-note/chord pool; wraps when the pool is shorter.
OCTAVE	Adds -2 to +2 octaves to the selected note.
VELOCITY	MIDI 0–127; zero is a rest. The velocity-mode menu sets key/step blending.
GATE	1–100 percent of the step or ratchet interval.
REPEAT	1–4 equally spaced note repetitions within a step.
CHANCE	0–100 percent chance of playing the step; zero is a rest.
TIE	Carries the note into the next step; repeated steps use their individual gates.

Draw values by dragging vertically across a row and horizontally across steps. Right-click or double-click a cell for exact numeric entry; scroll changes a cell by one. Clicking or painting the TIE row toggles ties. Each row has a playhead that follows its own length; cells beyond its length appear dimmed. For example, note length 5 and octave length 3 produce a 15-step combined cycle. The NOTE row uses held-note rank, rather than a recorded absolute melody.

TIME selects a musical division with SYNC, or seconds otherwise. LATCH keeps a released chord running until a new chord arrives. Turning latch off or stopping host transport clears latched notes. Pattern slots 1–8 store complete arp settings: choose a slot and SAVE, then select it later to recall. Empty slots do not change the pattern. DICE randomises all 32 steps. Chord maps are stored separately from these pattern slots. This version has one pattern playing at a time and manual slot selection, not an automatic song-chain mode.

MIDI OUT in the header sends generated notes, pitch/controller/pressure events and supported passthrough messages to the host in addition to playing the synth. Route its MIDI output to another instrument using your host's routing controls. Turning MIDI OUT off sends all-notes-off to prevent downstream stuck notes. MIDI output routing depends on the host. In Standalone, select a MIDI output device in the wrapper's audio/MIDI settings. Large SysEx messages are not forwarded. Internal processing preserves sample offsets and ratchet timing across buffer sizes.

Legacy mode and compatibility

Legacy retains the ten original modes, velocity modes, 16 steps, transpose, wrap, gate, swing, octaves and hold. The Step sequence factory sound uses it. Older logisth states select Legacy automatically when they have no Pattern engine parameter. ARP enable clears held notes; play fresh notes afterward. Pattern uses note-rank/octave/gate lanes instead of Legacy transpose/wrap and its shared gate. Legacy controls remain available in the Legacy view.

7. Effects rack

Signal order is fixed: Distortion Phaser Chorus EQ Reverb Delay Compressor Main. Click a name to edit it and use the adjacent square to bypass or enable it. All seven effects are global after the A/B mix.

Section	Controls and use
Distortion	Overdrive, Foldback, Clip, Decimate and Bitcrush; Amount and Dry/Wet set character and blend.

Section	Controls and use
Phaser	Six-stage stereo phasing with centre frequency, spread, L/R offset, width, LFO, feedback and Dry/Wet. Its frequency is a modulation destination.
Chorus	Delay, depth, rate, feedback, dual mode, width and Dry/Wet thicken a sound.
Equalizer	Bass and treble shelves with frequency controls and optional positive-gain compensation.
Reverb	Size, damping, predelay, width and Dry/Wet shape the room tail.
Delay	Independent left/right times or divisions, feedback, ping-pong, spread, width, low/high cut, smear and Dry/Wet.
Compressor	Stereo-linked threshold, ratio, attack and release control dynamics.

When SYNC is on, rate/time fields that support synchronization become division menus. Turning SYNC off exposes their independent Hz/time values again. Bypass can freeze a section's internal memory; a tail may resume when re-enabled. Delay memory is limited to 16 seconds per channel, including very slow synced divisions. Compressor attack/release and some effect delays use milliseconds; envelope times use seconds internally. The panel displays suitable units.

8. Mixer and performance

MIX A and MIX B set the part levels; MAIN controls output after all effects. Independent L/R meters show post-effects output. There is no hard output limiter: loud patches, resonance and effects may exceed 0 dBFS. Reduce MAIN as needed.

POLY sets up to 16 simultaneous notes. Each note may contain several unison voices. SOLO isolates A or B. MONO / LEGATO uses last-note priority and returns to the previous held note when the newest one is released. POLY is disabled in mono mode. GLIDE sets portamento time; Normal mode glides for overlapping notes, while Slide also permits glide between separate notes.

BEND / st sets pitch-wheel range in semitones. The on-screen BEND wheel springs back to the centre after dragging. MOD remains where you set it. Route Mod Wheel through a Misc block to hear its effect. MIDI pitch bend and modulation messages also update these controls. The keyboard covers five octaves.

9. Presets, automation and project recall

The ten factory sounds are Init / analog, Wide supersaw, Sub bass, Slow atmosphere, Pulse pluck, Step sequence, Morph glass, Sample keys, Grain cloud and Chord pulse. Choose a sound in SOUND LIBRARY or use the arrow buttons. Loading a sound replaces the current patch; save the project first if you want to retain edits. The preset selection is a factory sound selector, not a user bank manager. Use your DAW's plug-in preset/state system to save patches.

All 514 audio parameters are available for host automation. A/B view changes, rack selection and theme do not alter their values. The theme is saved as UI metadata rather than an additional audio parameter. DAW state contains all audio parameters, modulation routes and the 16-step pattern. Reopening an instance restores its stored theme. The first 248 parameter IDs remain unchanged, with 266 new IDs appended. The destination menus now have six additional entries. Projects restore stored patch values; review destination automation recorded as normalised values in older hosts, since expanding a choice list can change its normalised positions.

There is no Sylenth1 bank import or MIDI Learn menu. Legacy prototype states receive a best-effort migration of filter and modulation settings; changes to DSP mean that identical sound is not guaranteed.

10. Sound-design examples

For a wide lead, load Wide supersaw. Raise unison detune gradually, open shared CUTOFF, and add a little Delay. Use MONO / LEGATO with a short GLIDE for connected melodies. Lower MAIN if the combined voices become too loud.

For an envelope sweep, load Init / analog, lower Filter A cutoff, select Cutoff A as Mod Env 1's first destination, and add a small positive amount. Set attack near zero, moderate decay, low sustain and short release. Increase the amount until the filter opens sufficiently at the start of each note.

For vibrato, choose Mod Wheel in a Misc source block and route it to LFO 1 Gain. Set LFO 1 to a sine wave in FREE mode at a few Hz, set its base Gain to zero, then route LFO 1 to Pitch A+B with a small amount. The wheel opens the vibrato depth. Keep amounts low to avoid large pitch excursions.

For a sequence, load Step sequence and hold a note. Adjust PITCH and VELOCITY, set a zero velocity for a rest, and enable HOLD to tie steps. Select 09–16 and increase LENGTH to use the second half. Automate the shared filter cutoff to move both parts together without replacing their local settings.

11. Troubleshooting and technical notes

High pitches in Sampler/Granular use interpolated resampling and may alias. Wavetable mip bands reduce aliasing, but phase warp can introduce new harmonics. These engines are independent implementations, not models of commercial synths. Dense granular patches with high unison/polyphony require more CPU. Imported material can be loud; use moderate levels. An import failure leaves the previous material intact. A source starts again on the next note or engine/material change.

Release 0.4.0 introduces these engines and the new chord/pattern workflow. Native archives are published only after their platform builds and tests pass. See BUILD_INFO.json in each archive for source revision and validation results.

No sound: check the audio/MIDI device or DAW routing, oscillator Voices and Volume, filter input, envelope sustain, Mix levels and MAIN. Check SOLO and whether the arp needs fresh notes. A long attack or low cutoff can hide a note.

No modulation: check the source is active, destination is not None, and amount is nonzero. The mod wheel needs an assigned route. A source can be zero during part of a note; Step Velocity needs an active arp gate.

Missing control: A/B selects the part, a rack name selects effect controls, and SYNC/FREE selects time versus rhythmic controls. RETRIG enables phase editing. The second group of arp steps is under 09–16.

High CPU: reduce unison voices, polyphony or active effects. Some abrupt changes to routing, waveforms and delay times can click. Selected continuous controls are smoothed, but not every transition is crossfaded.

Build validation includes audio at 44.1/48/96 kHz, filters, modulation, arp timing, state restoration, preset output and editor bindings/layout/theme persistence. Linux tests additionally load the actual CLAP and VST3 binaries. A VST3 factory smoke test does not replace testing in your particular DAW. Native macOS tests run on Intel; Universal architecture checks establish that arm64 code is present, not that it has been auditioned on Apple Silicon hardware.

The filter is an independent SVF design; reverb uses JUCE Reverb. Modulation scales and some sequencer behaviour are implementation choices. No sample-exact comparison with Sylenth1 has been performed. Third-party font notices are included in assets/fonts. The original Sylenth1 manual and graphics are not redistributed.